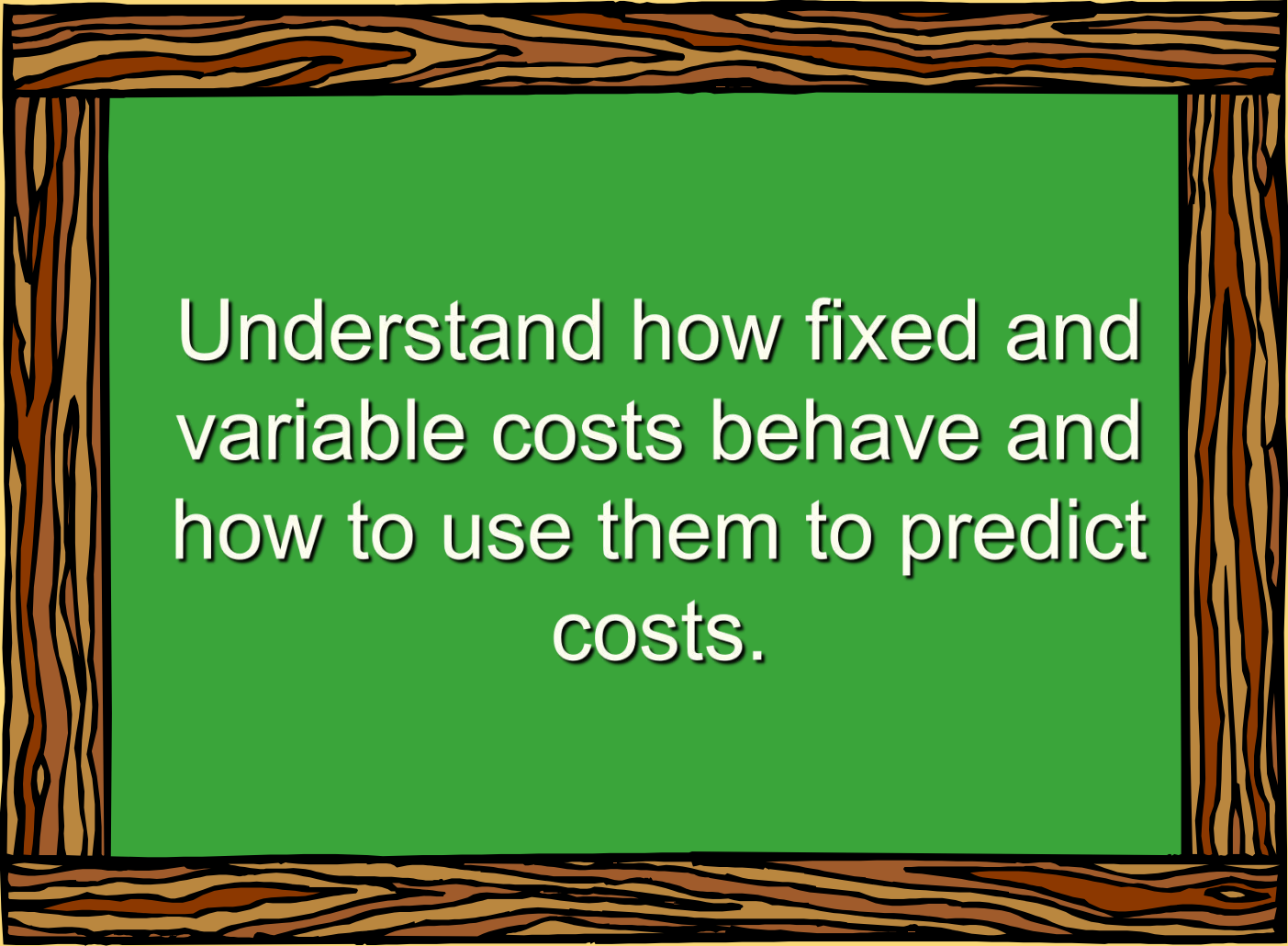


Cost Behavior: Analysis and Use

Chapter Five



Learning Objective 1



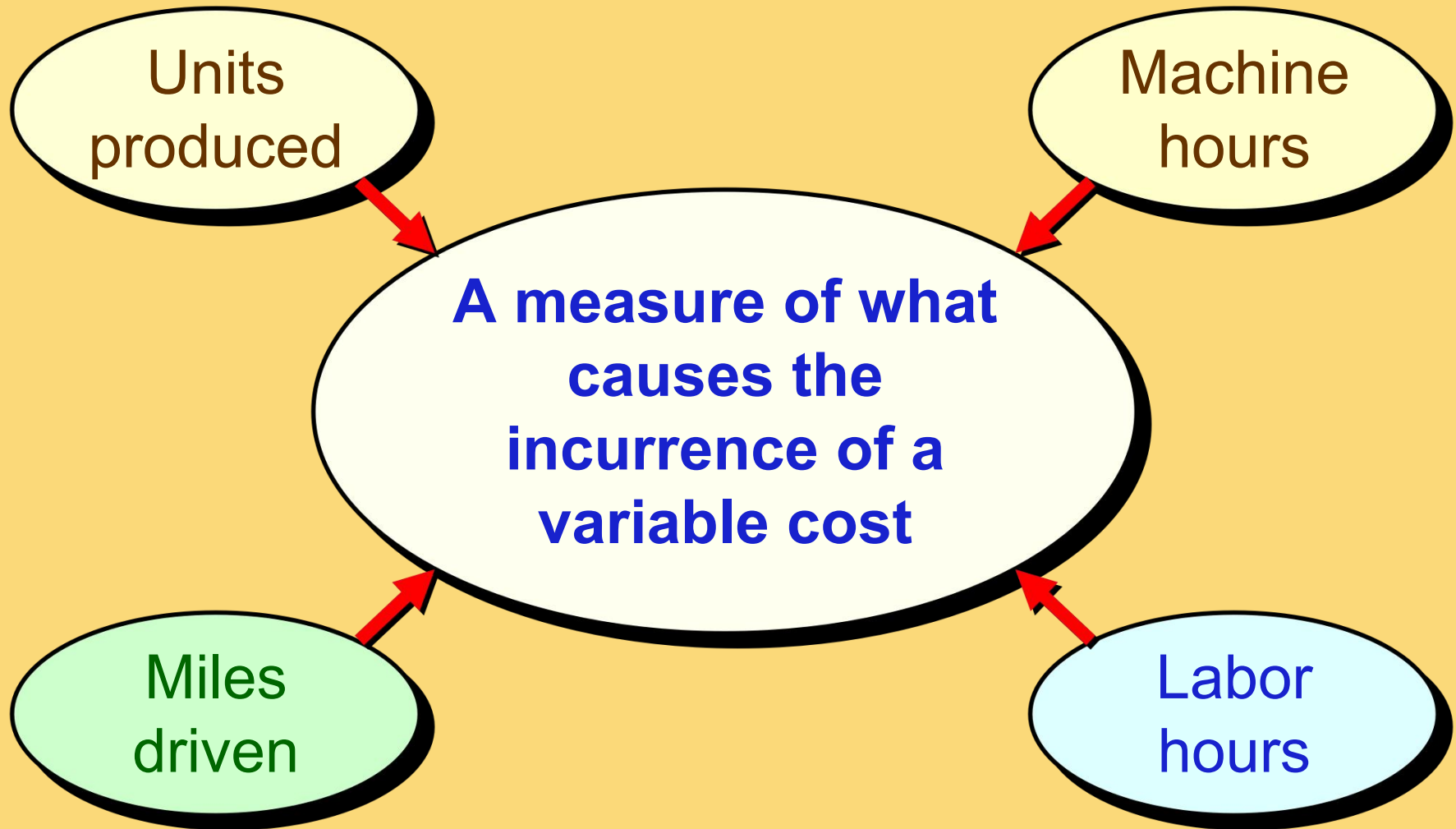
Understand how fixed and variable costs behave and how to use them to predict costs.

Types of Cost Behavior Patterns

Recall the summary of our cost behavior discussion from an earlier chapter.

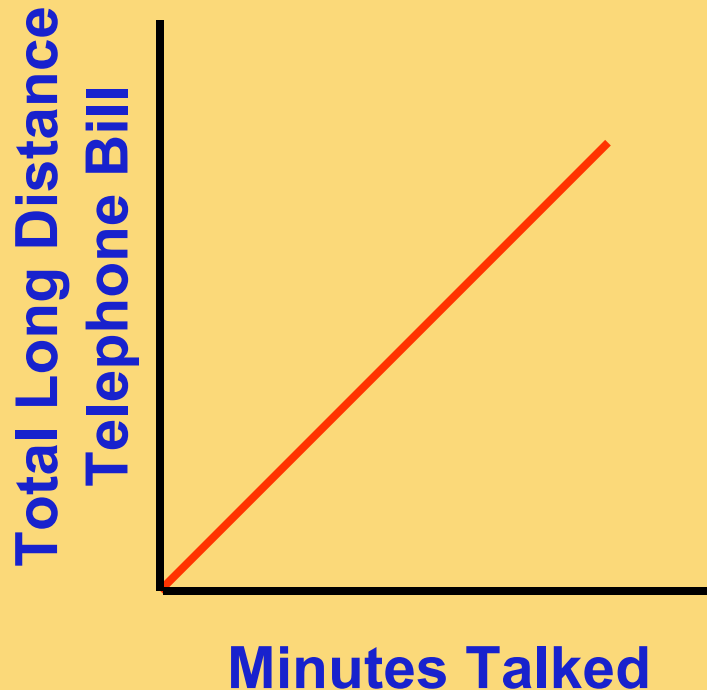
Summary of Variable and Fixed Cost Behavior		
Cost	In Total	Per Unit
Variable	Total variable cost is proportional to the activity level within the relevant range.	
Fixed	Total fixed cost remains the same even when the activity level changes within the relevant range.	Fixed cost per unit goes down as activity level goes up.

The Activity Base (Cost driver)



The Activity Base Example

A variable cost is a cost whose total dollar amount varies in direct proportion to changes in the activity level. Your total long distance telephone bill is based on how many minutes you talk.



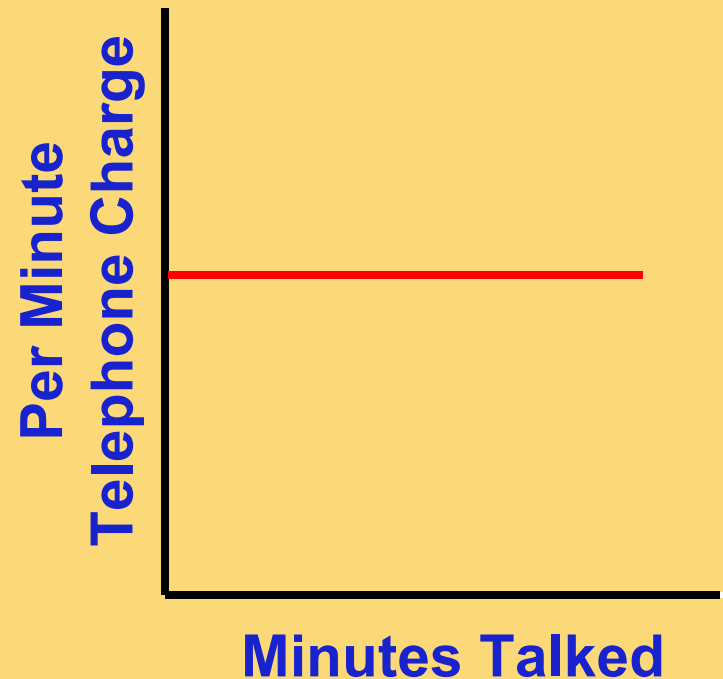
Types of Cost Behavior Patterns

Recall the summary of our cost behavior discussion from an earlier chapter.

Summary of Variable and Fixed Cost Behavior		
Cost	In Total	Per Unit
Variable	Total variable cost is proportional to the activity level within the relevant range.	Variable cost per unit remains the same over wide ranges of activity.
Fixed	Total fixed cost remains the same even when the activity level changes within the relevant range.	Fixed cost per unit goes down as activity level goes up.

Variable Cost Per Unit Example

A variable cost remains constant if expressed on a per unit basis. The cost per minute talked is constant. For example, 20 cents per minute.



Extent of Variable Costs

The proportion of variable costs *differs* across organizations. For example . . .

A public utility with large investments in equipment will tend to have *fewer* variable costs.

A manufacturing company will often have *many* variable costs.

Service companies have diverse cost structures.

A merchandising company usually will have a *high proportion of variable costs*, like cost of sales.

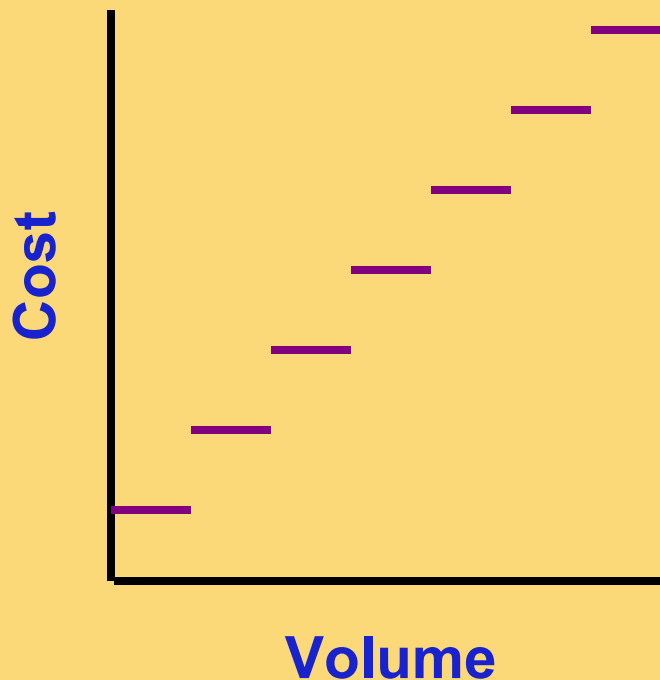
Examples of Variable Costs

1. *Merchandising companies* – cost of goods sold.
2. *Manufacturing companies* – direct materials, direct labor, and variable overhead.
3. *Merchandising and manufacturing companies* – commissions, shipping costs, and clerical costs, such as invoicing.
4. *Service companies* – supplies, travel, and clerical.



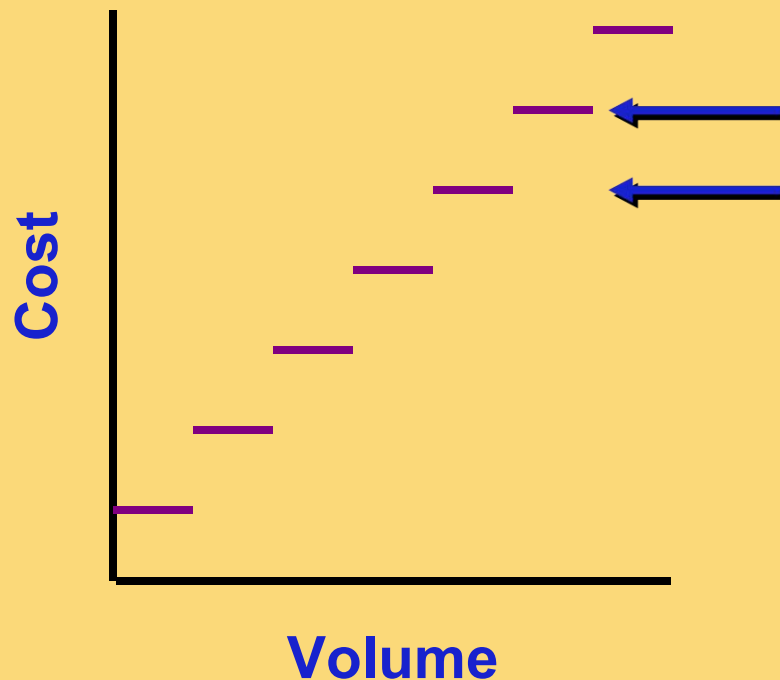
Step-Variable Costs

A resource that is obtainable only in large chunks (such as maintenance workers) and whose costs increase or decrease only in response to fairly wide changes in activity is known as a **step-variable cost**.

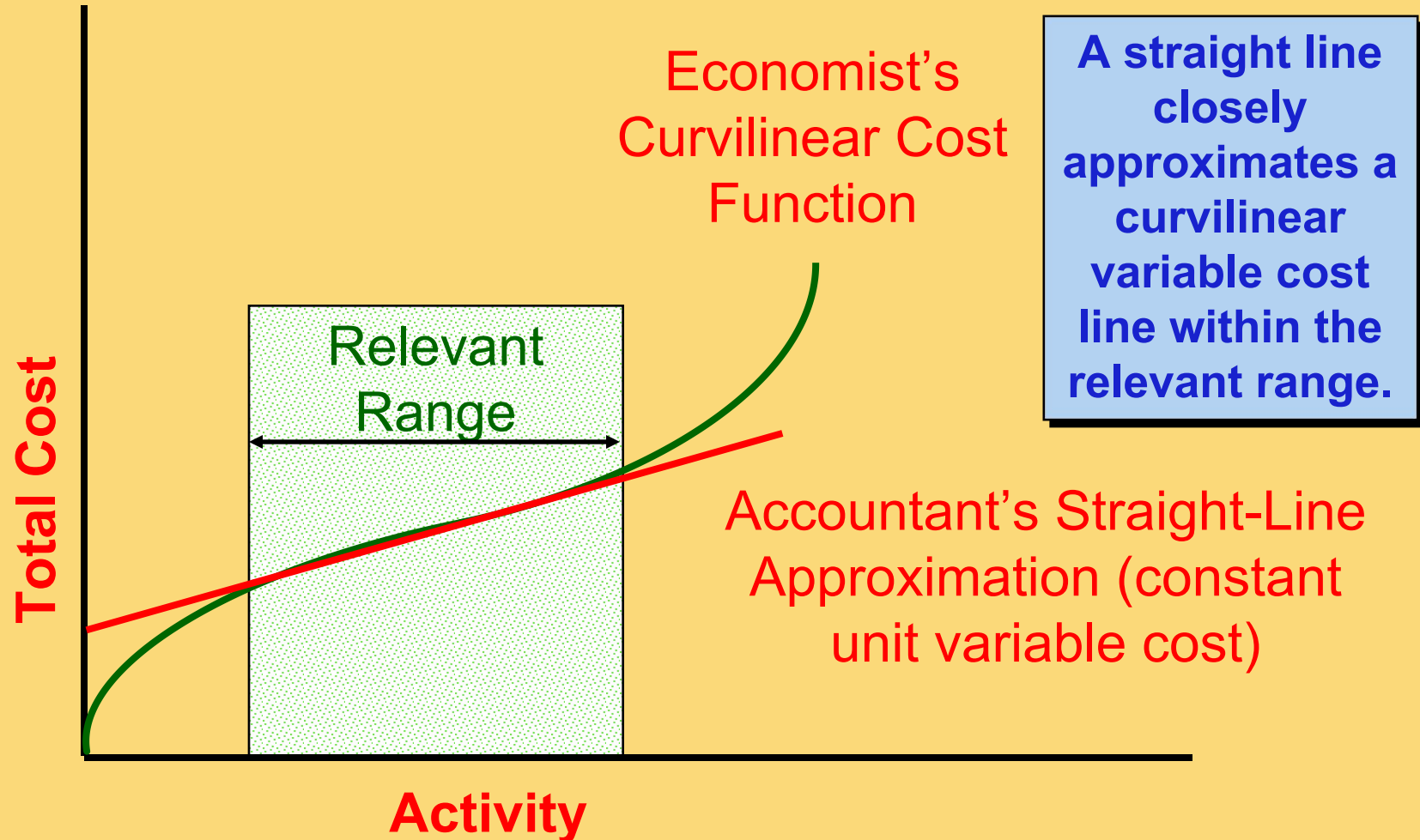


Step-Variable Costs

Only fairly wide changes in the activity level will cause a change in the number of maintenance workers employed



The Linearity Assumption and the Relevant Range



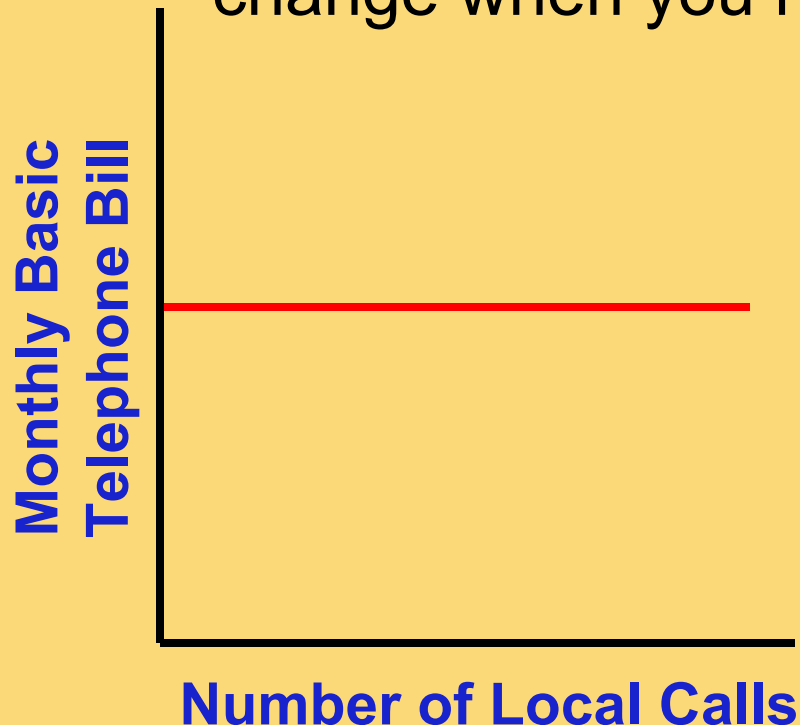
Types of Cost Behavior Patterns

Let's look at fixed cost behavior on the next screens.

Summary of Variable and Fixed Cost Behavior		
Cost	In Total	Per Unit
Variable	Total variable cost is proportional to the activity level within the relevant range.	Variable cost per unit remains the same over wide ranges of activity.
Fixed	Total fixed cost remains the same even when the activity level changes within the relevant range.	Fixed cost per unit goes down as activity level goes up.

Total Fixed Cost Example

A fixed cost is a cost whose total dollar amount remains constant as the activity level changes. Your monthly basic telephone bill is probably fixed and does not change when you make more local calls.



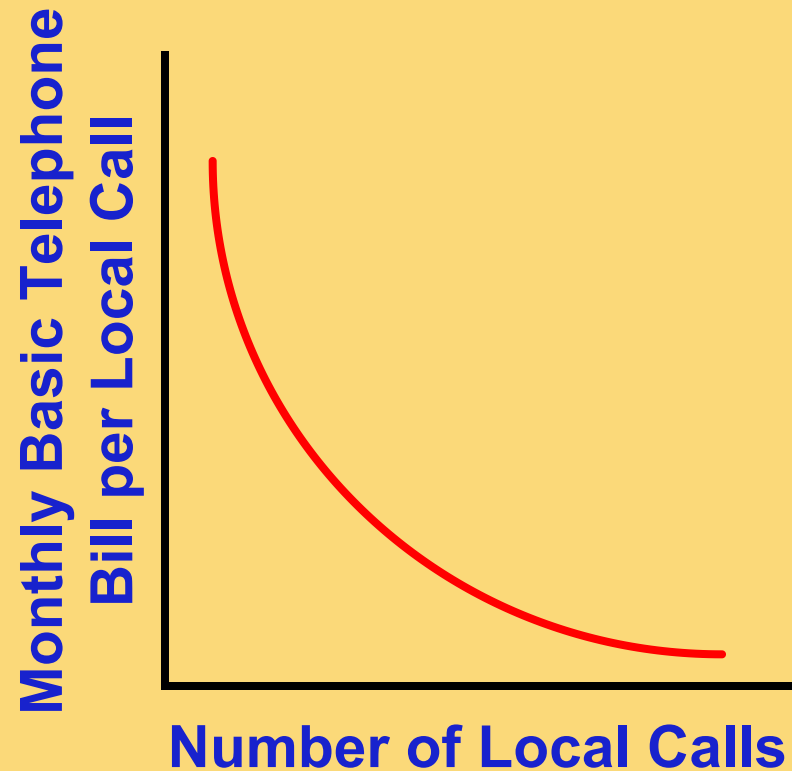
Types of Cost Behavior Patterns

Recall the summary of our cost behavior discussion from an earlier chapter.

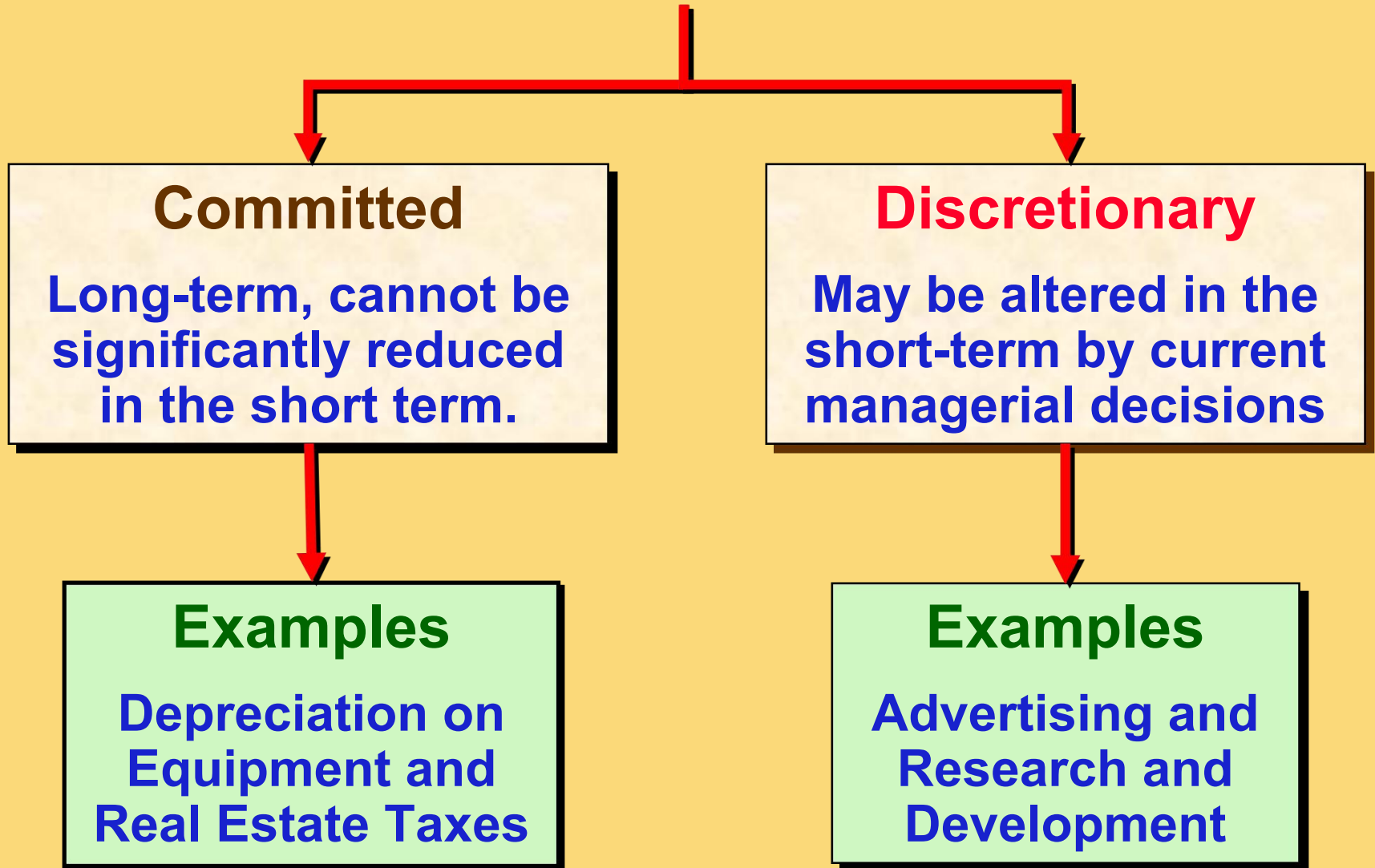
Summary of Variable and Fixed Cost Behavior		
Cost	In Total	Per Unit
Variable	Total variable cost is proportional to the activity level within the relevant range.	Variable cost per unit remains the same over wide ranges of activity.
Fixed	Total fixed cost remains the same even when the activity level changes within the relevant range.	Fixed cost per unit goes down as activity level goes up.

Fixed Cost Per Unit Example

Average fixed costs per unit decrease as the activity level increases. The fixed cost per local call decreases as more local calls are made.



Types of Fixed Costs



The Trend Toward Fixed Costs

The trend in many industries is toward *greater fixed costs* relative to variable costs.

As machines take over many mundane tasks previously performed by humans, “*knowledge workers*” are demanded for their minds rather than their muscles.

Knowledge workers tend to be salaried, highly-trained and difficult to replace. The cost to compensate these valued employees is *relatively fixed* rather than variable.

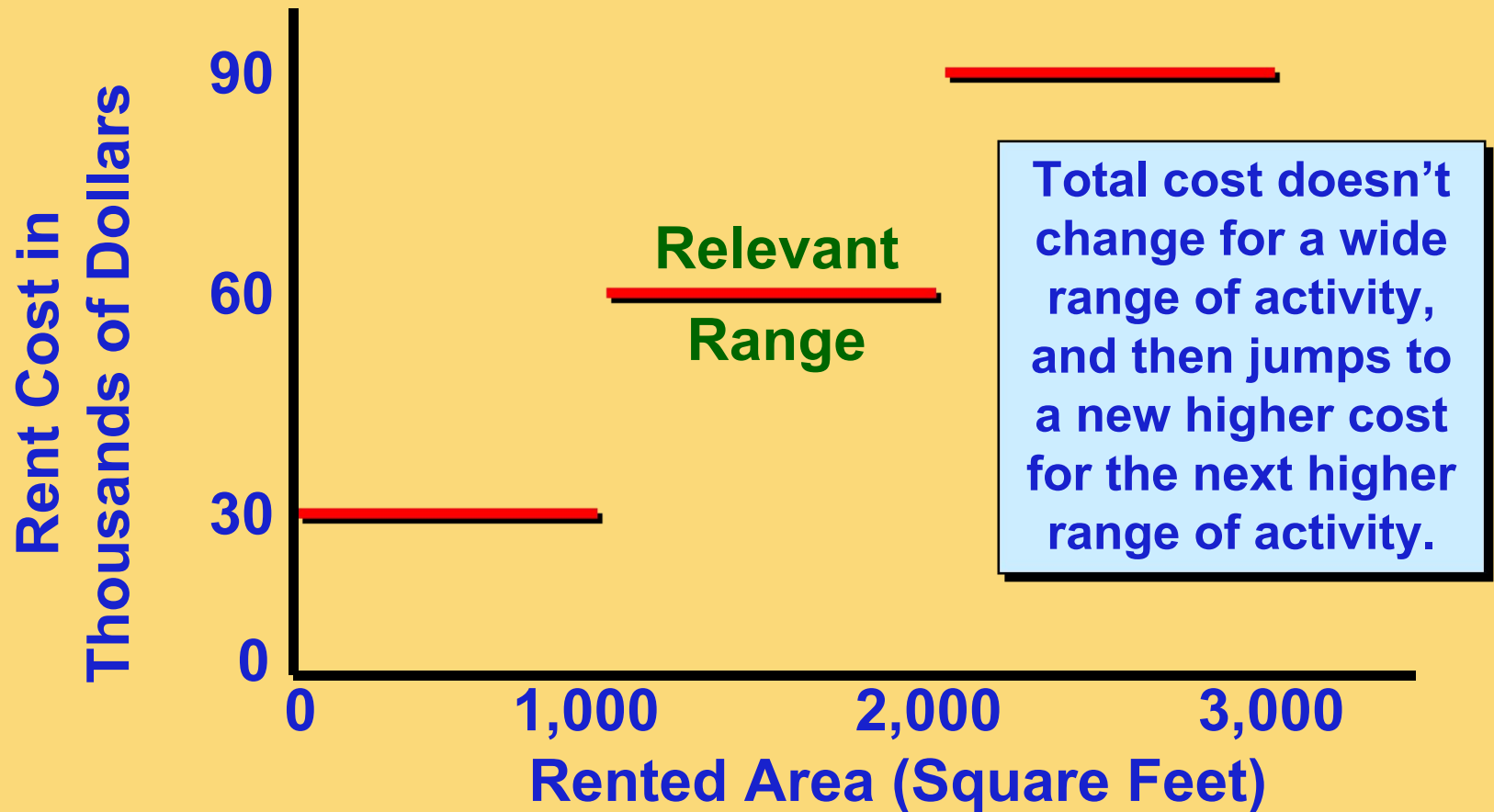
Is Labor a Variable or a Fixed Cost?

The behavior of wage and salary costs can *differ across countries*, depending on labor regulations, labor contracts, and custom.

In *France*, *Germany*, *China*, and *Japan*, management has little flexibility in adjusting the size of the labor force. Labor costs are more fixed in nature.

In the *United States* and the *United Kingdom*, management has greater latitude. Labor costs are more variable in nature.

Fixed Costs and Relevant Range



Fixed Costs and Relevant Range

The relevant range of activity for a fixed cost is the range of activity over which the graph of the cost is flat.

Example: Office space is available at a rental rate of \$30,000 per year in increments of 1,000 square feet. As the business grows, more space is rented, increasing the total cost.



Fixed Costs and Relevant Range



**How does this
type of fixed cost
differ from a step-
variable cost?**



**Step-variable costs
can be adjusted
more quickly and . . .**

**The width of the
activity steps is
much wider for the
fixed cost.**

Quick Check ✓

Which of the following statements about cost behavior are true?

1. **Fixed costs per unit vary with the level of activity.**
2. **Variable costs per unit are constant within the relevant range.**
3. **Total fixed costs are constant within the relevant range.**
4. **Total variable costs are constant within the relevant range.**

Quick Check ✓

Which of the following statements about cost behavior are true?

- 1. Fixed costs per unit vary with the level of activity.**
- 2. Variable costs per unit are constant within the relevant range.**
- 3. Total fixed costs are constant within the relevant range.**
4. Total variable costs are constant within the relevant range.

A Case

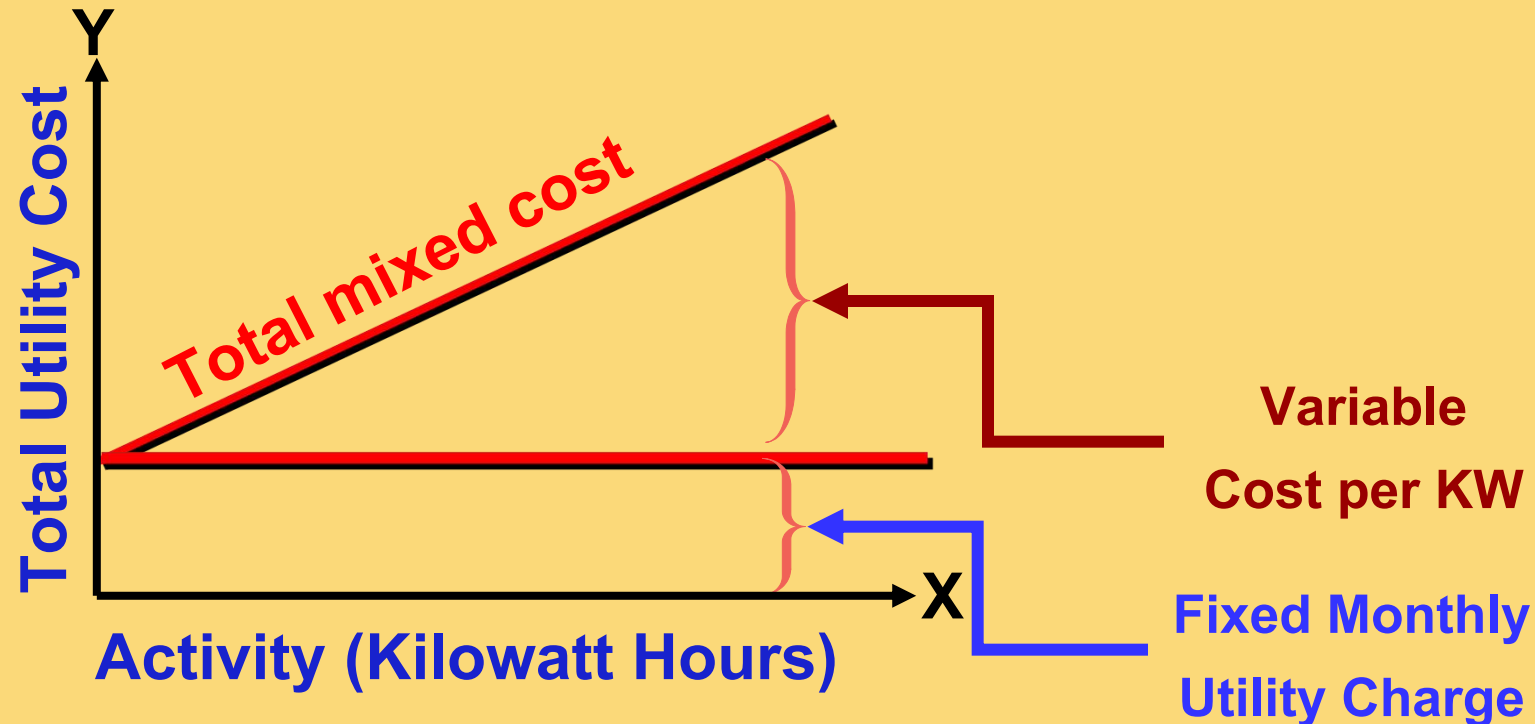
- Jackson is owned and operated by Jill and David. The company provides guided tours to remote areas, with the llamas carrying the baggage for the multiday treks. Jill and David operate out of their ranch. All of food is provided as well as tents and sleeping pads. Based on the number of guests on a trip, Jill and David will decide how many llamas will go on the trip and how many will remain on the ranch. Llamas are transported to the trailhead in a special trailer. The company has a number of costs, some of which are listed on the next slide.

Cost	Variable or Fixed	Activity Base
Food and beverage costs		
Truck and trailer operating costs		
Guide wages		
Costs of providing tents		
Cost of feeding llamas		
Property taxes		

Cost	Variable or Fixed	Activity Base
Food and beverage costs	Variable	the Number of guests and the length of the trip in days
Truck and trailer operating costs	Variable	the Number of miles to the trailhead
Guide wages	Step Variable	
Costs of providing tents	Variable	the Number of guests and the length of the trip in days
Cost of feeding llamas	Variable	the Number of llamas on the ranch
Property taxes	Fixed	

Mixed Costs

A mixed cost has both fixed and variable components. Consider the example of utility cost.

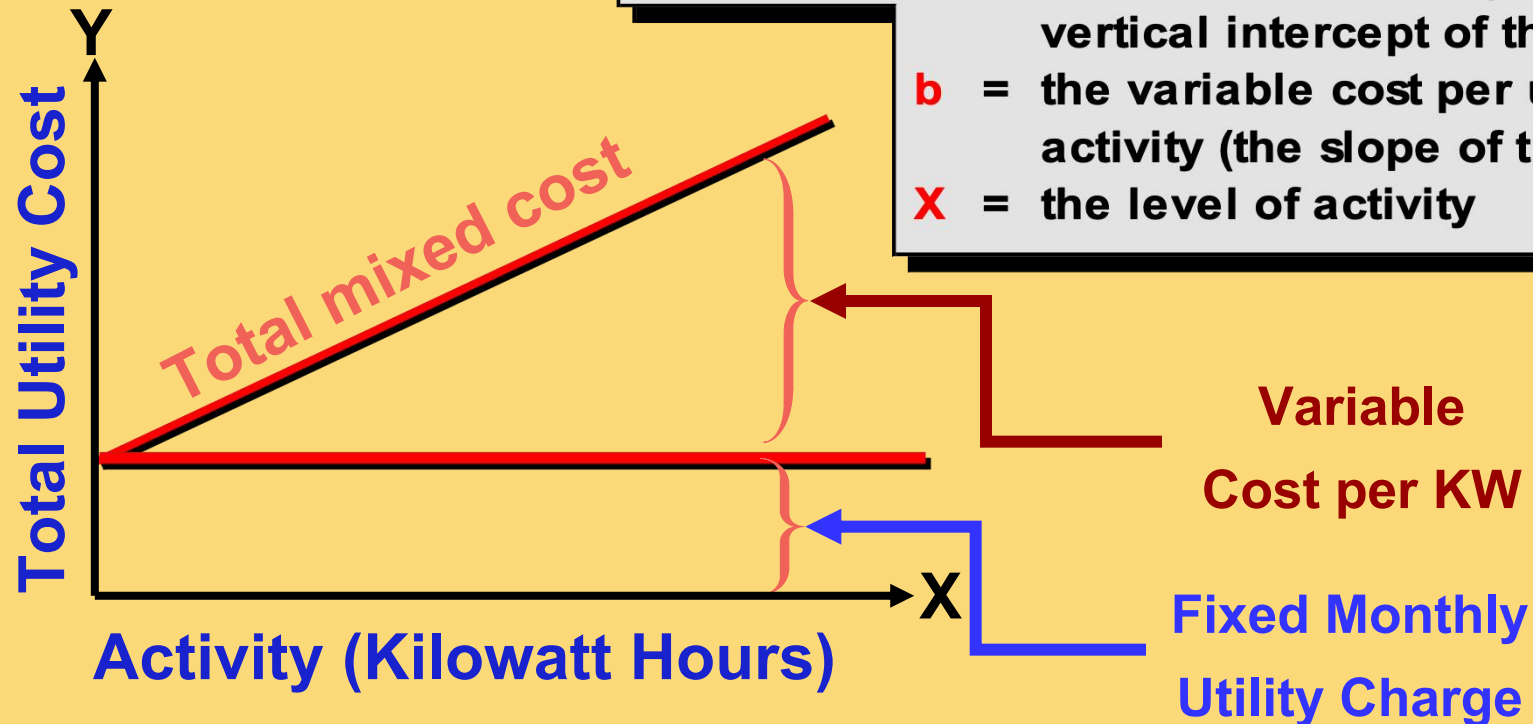


Mixed Costs

The total mixed cost line can be expressed as an equation: $Y = a + bX$

Where:

- Y = the total mixed cost
- a = the total fixed cost (the vertical intercept of the line)
- b = the variable cost per unit of activity (the slope of the line)
- X = the level of activity



Mixed Costs Example

If your fixed monthly utility charge is \$40, your variable cost is \$0.03 per kilowatt hour, and your monthly activity level is 2,000 kilowatt hours, what is the amount of your utility bill?

$$Y = a + bX$$

$$Y = \$40 + (\$0.03 \times 2,000)$$

$$Y = \$100$$

Analysis of Mixed Costs

Account Analysis and the Engineering Approach

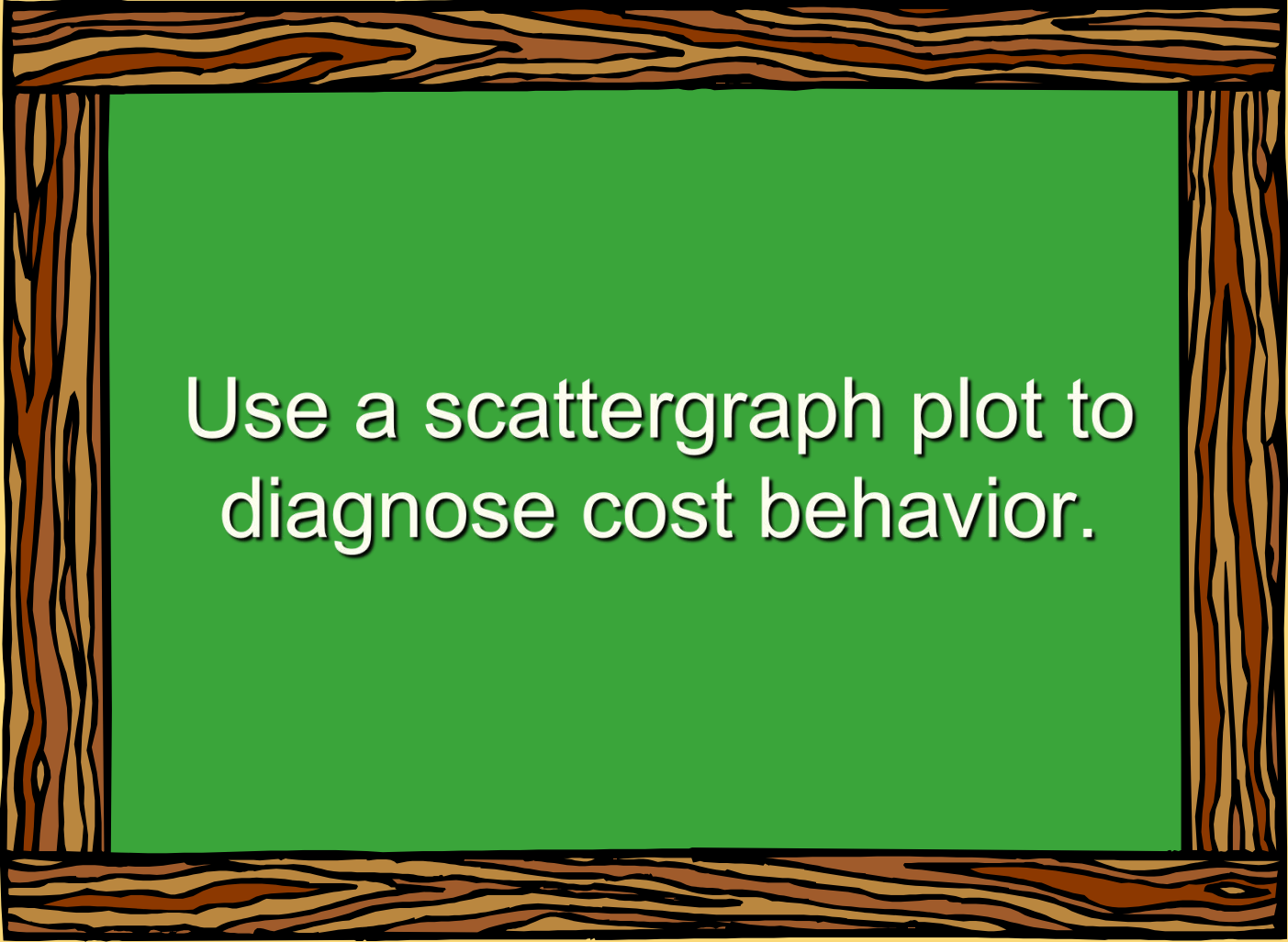


Each account is classified as either variable or fixed based on the analyst's knowledge of how the account behaves.



Cost estimates are based on an evaluation of production methods, and material, labor and overhead requirements.

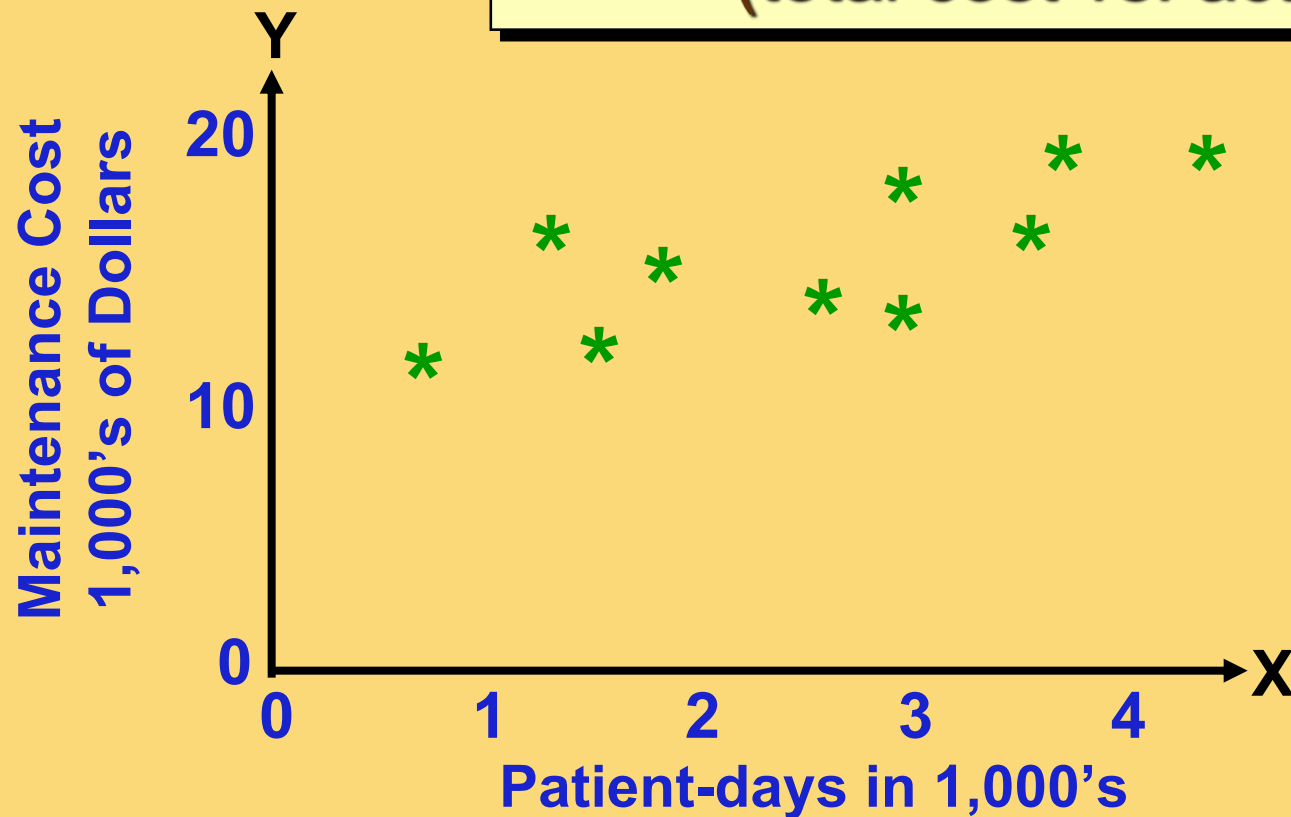
Learning Objective 2



Use a scattergraph plot to
diagnose cost behavior.

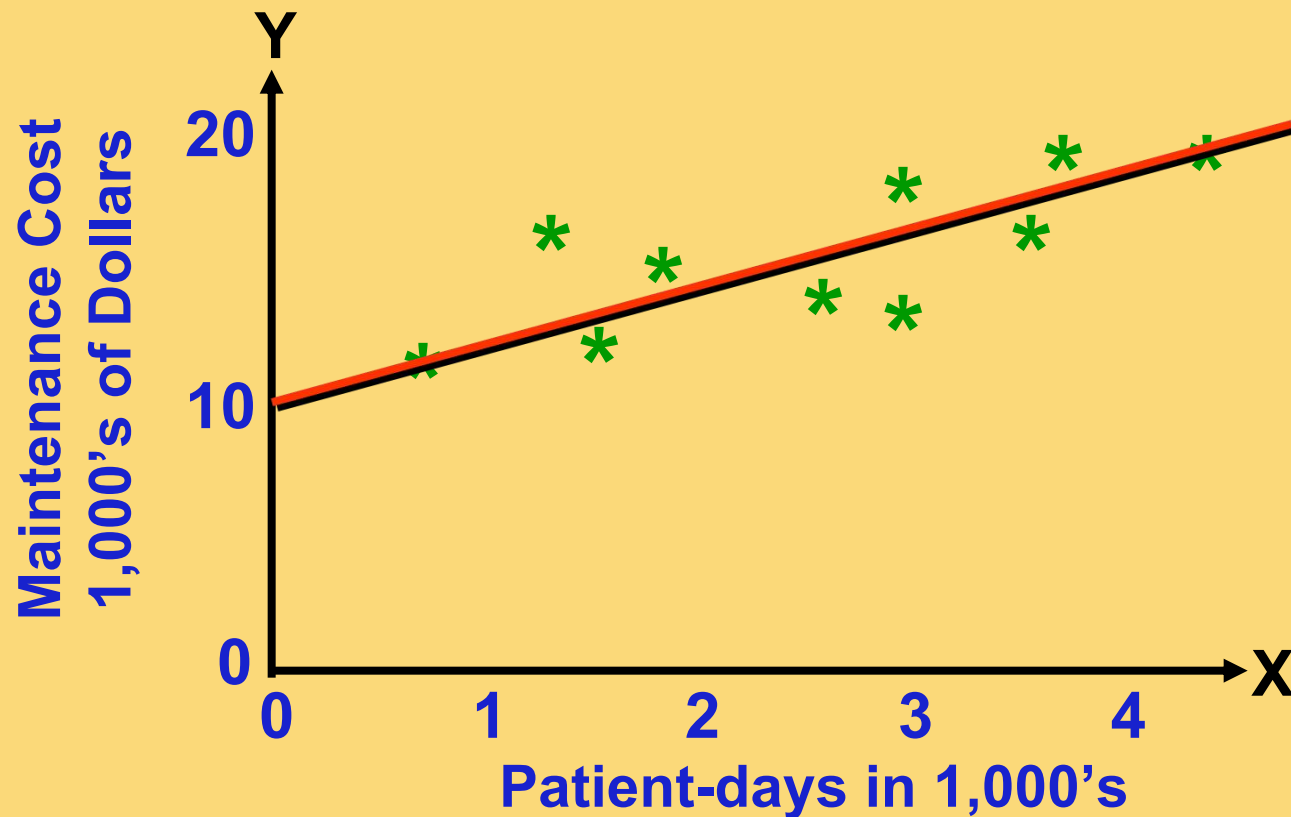
The Scattergraph Method

Plot the data points on a graph
(total cost vs. activity).



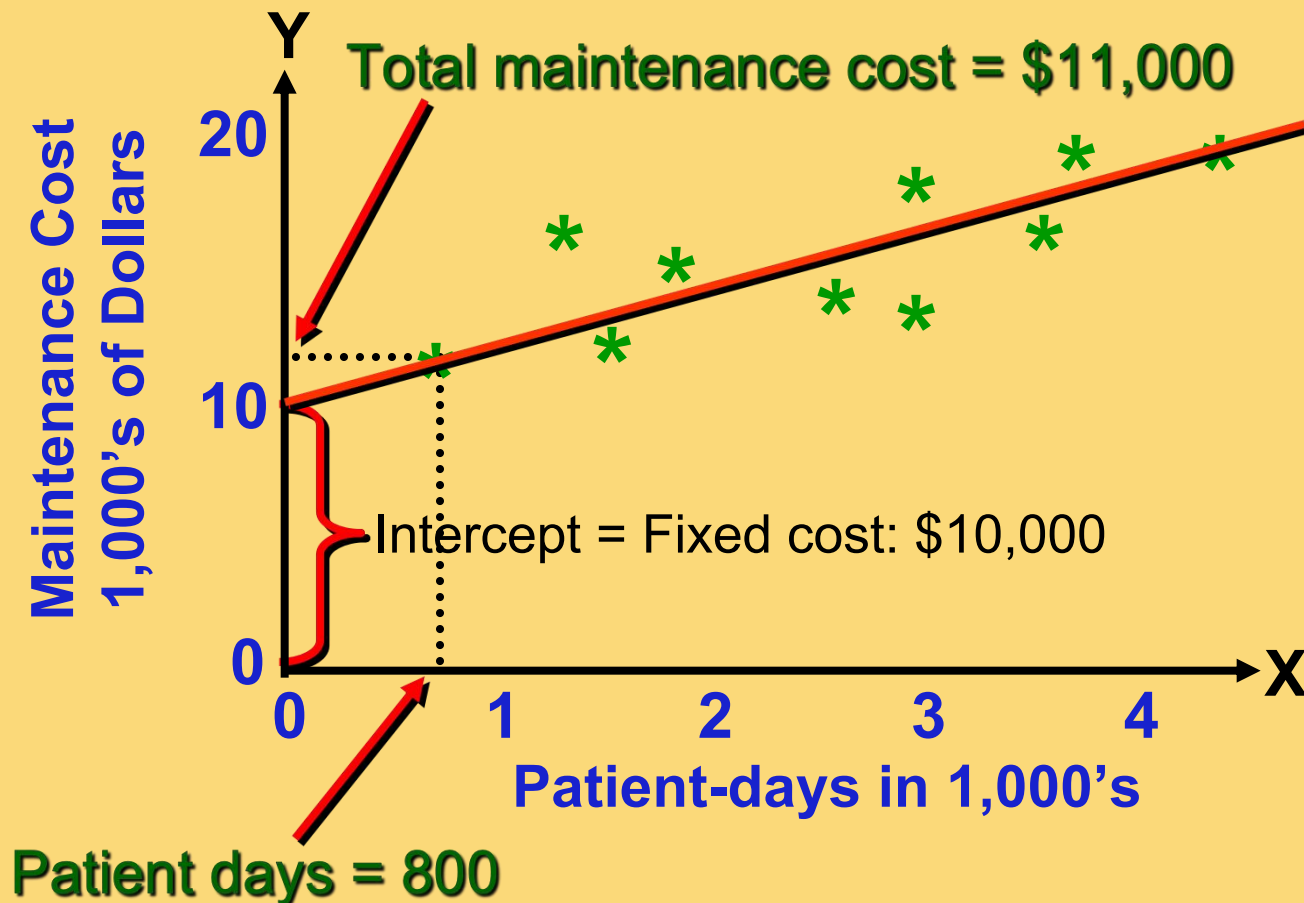
The Scattergraph Method

Draw a line through the data points with about an equal numbers of points above and below the line.



The Scattergraph Method

Use one data point to estimate the total level of activity and the total cost.



The Scattergraph Method

Make a quick estimate of variable cost per unit and determine the cost equation.

Total maintenance at 800 patients	\$ 11,000
Less: Fixed cost	10,000
Estimated total variable cost for 800 patients	<u>\$ 1,000</u>

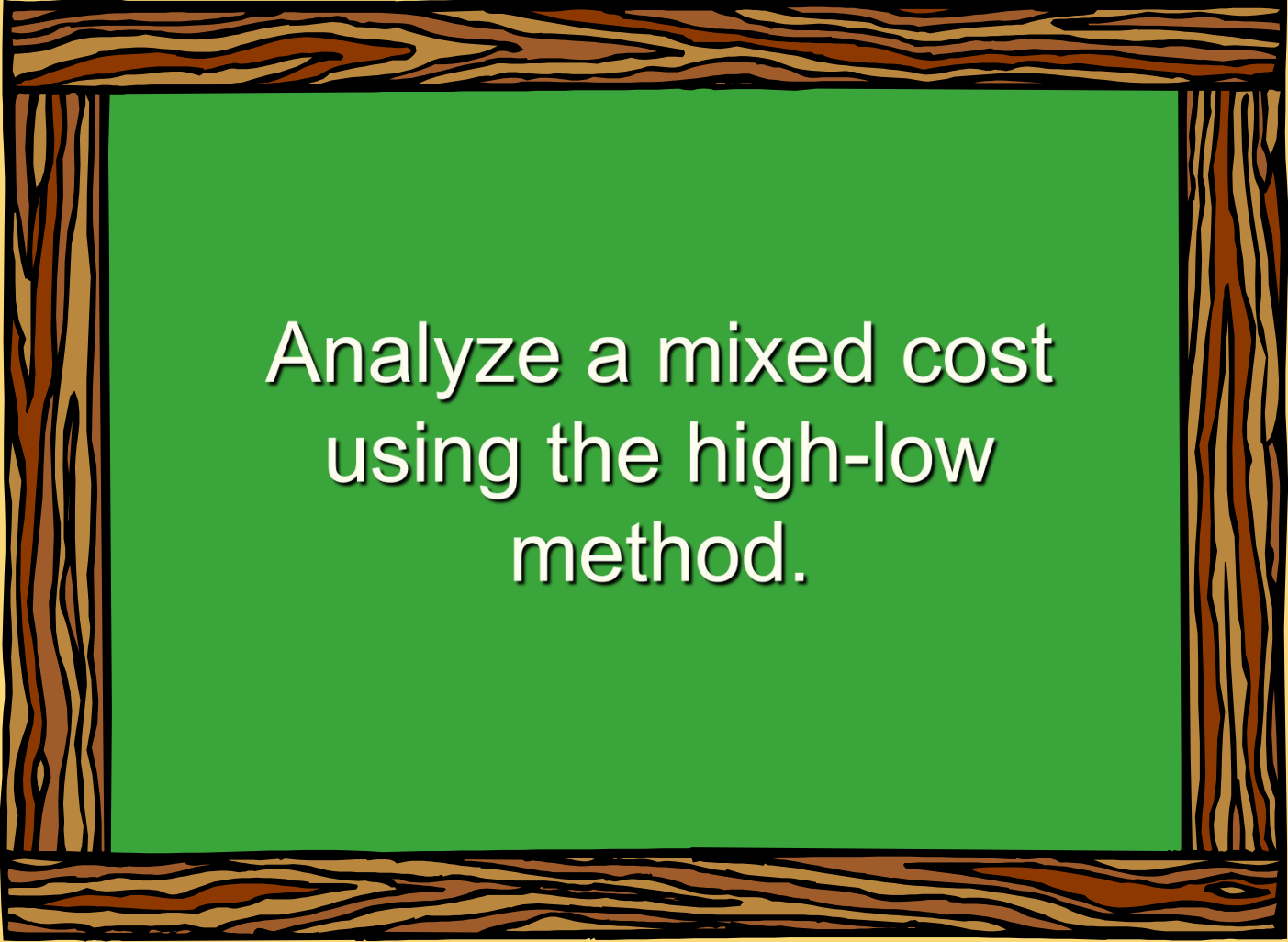
$$\text{Variable cost per unit} = \frac{\$1,000}{800} = \$1.25/\text{patient-day}$$

$$Y = \$10,000 + \$1.25X$$

Total maintenance cost

Number of patient days

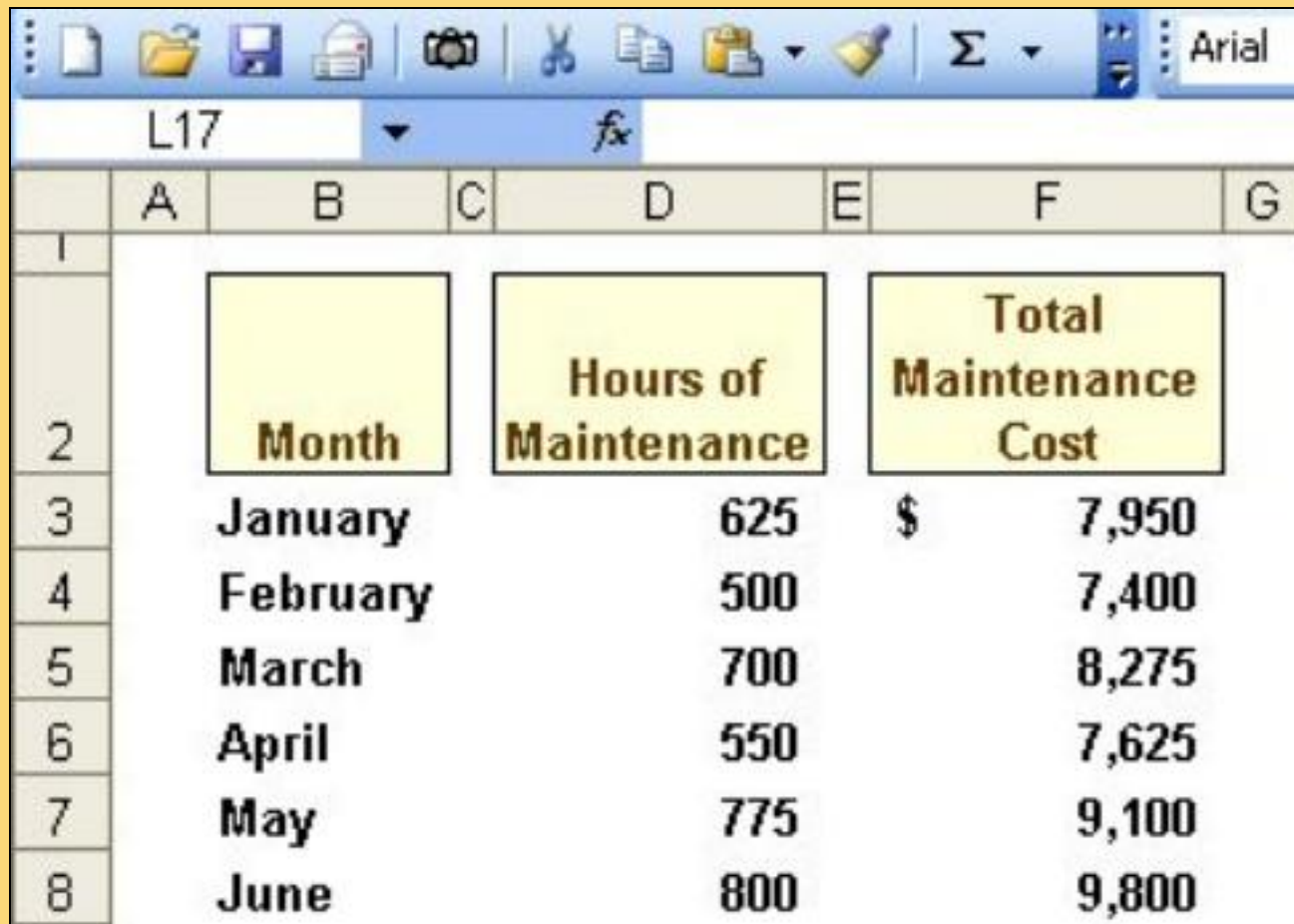
Learning Objective 3



Analyze a mixed cost
using the high-low
method.

The High-Low Method

Assume the following hours of maintenance work and the total maintenance costs for six months.



The image shows a screenshot of an Excel spreadsheet. The formula bar at the top displays 'L17' and a function icon. The spreadsheet has columns labeled A through G and rows numbered 1 through 8. The data is organized as follows:

	A	B	C	D	E	F	G
1							
2		Month		Hours of Maintenance		Total Maintenance Cost	
3		January		625	\$	7,950	
4		February		500		7,400	
5		March		700		8,275	
6		April		550		7,625	
7		May		775		9,100	
8		June		800		9,800	

The High-Low Method

M19						
	A	B	C	D	E	F
1						
2						
3						
4						
5						
6						
7						
8						
9						
10						
11						
12						

Month	Hours of Maintenance	Total Maintenance Cost
January	625	\$ 7,950
February	500	7,400
March	700	8,275
April	550	7,625
May	775	9,100
June	800	9,800
High Level	800	\$ 9,800
Low Level	500	7,400
Change	300	\$ 2,400

The *variable cost per hour* of maintenance is equal to the change in cost divided by the change in hours.

$$\frac{\$2,400}{300} = \$8.00/\text{hour}$$

The High-Low Method

	N21						
	A	B	C	D	E	F	G
1							
2		Month	Hours of Maintenance			Total Maintenance Cost	
9		High Level	800		\$	9,800	
10		Low Level	500			7,400	
11		Change	300		\$	2,400	
12							

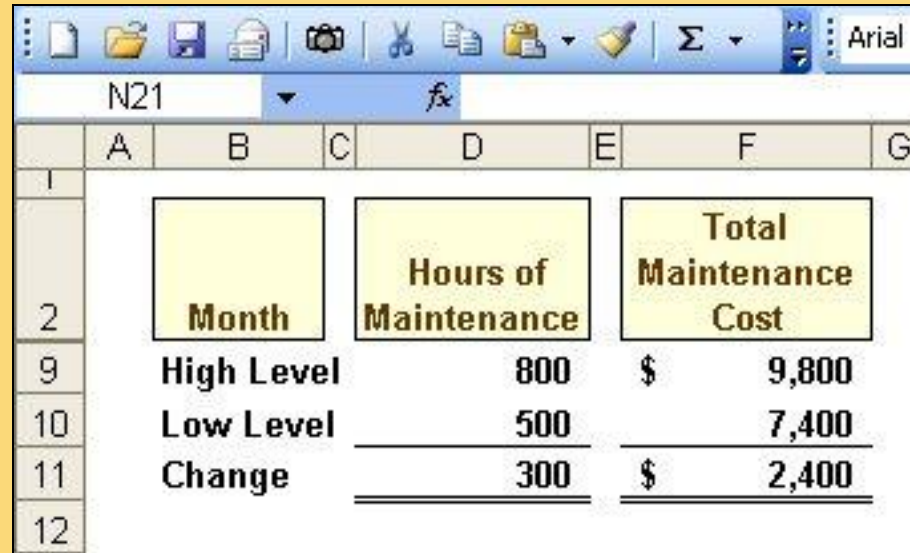
Total Fixed Cost = Total Cost – Total Variable Cost

Total Fixed Cost = \$9,800 – (\$8/hour × 800 hours)

Total Fixed Cost = \$9,800 – \$6,400

Total Fixed Cost = \$3,400

The High-Low Method



The screenshot shows an Excel spreadsheet with the following data:

	A	B	C	D	E	F	G
1							
2		Month		Hours of Maintenance		Total Maintenance Cost	
9		High Level		800		\$ 9,800	
10		Low Level		500		7,400	
11		Change		300		\$ 2,400	
12							

The Cost Equation for Maintenance

$$Y = \$3,400 + \$8.00X$$

Quick Check ✓

Sales salaries and commissions are \$10,000 when 80,000 units are sold, and \$14,000 when 120,000 units are sold. Using the high-low method, what is the **variable** portion of sales salaries and commission?

- a. \$0.08 per unit
- b. \$0.10 per unit
- c. \$0.12 per unit
- d. \$0.125 per unit

Quick Check ✓

Sales salaries and commissions are \$10,000 when 80,000 units are sold, and \$14,000 when 120,000 units are sold. Using the high-low method, what is the **fixed** portion of sales salaries and commissions?

- a. \$ 2,000
- b. \$ 4,000
- c. \$10,000
- d. \$12,000

Least-Squares Regression Method

A method used to analyze mixed costs if a scattergraph plot reveals an approximately linear relationship between the X and Y variables.

This method uses *all* of the data points to estimate the fixed and variable cost components of a mixed cost.

$$\begin{array}{r} 10784.36 \\ 5 \overline{) 53913.72} \\ \underline{2719372} \end{array}$$

The goal of this method is to fit a straight line to the data that *minimizes the sum of the squared errors*.

Least-Squares Regression Method

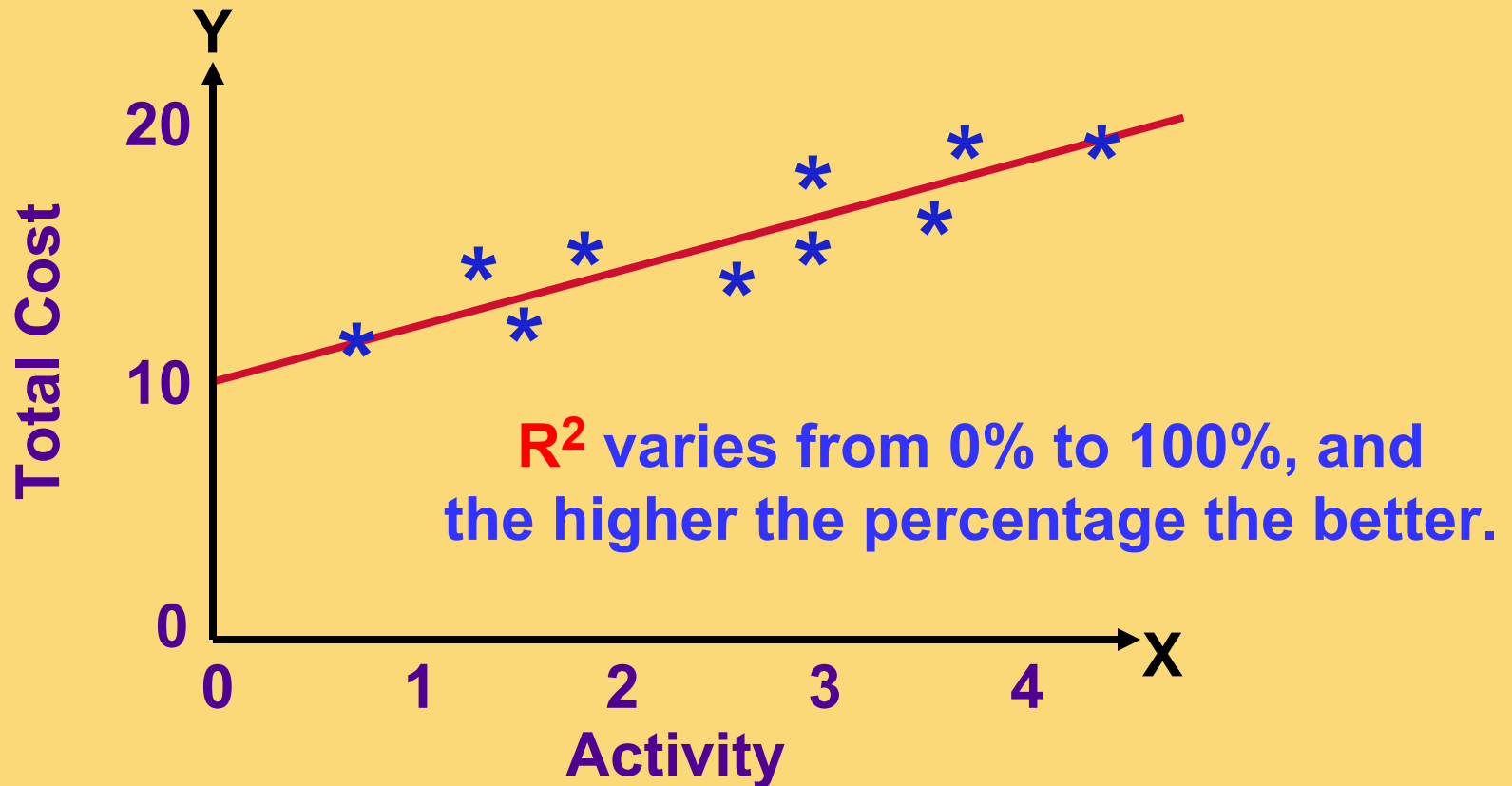
- Software can be used to fit a regression line through the data points.
- The cost analysis objective is the same: $Y = a + bX$



Least-squares regression also provides a statistic, called the R^2 , which is a measure of the goodness of fit of the regression line to the data points.

Least-Squares Regression Method

R^2 is the percentage of the variation in total cost explained by the activity.



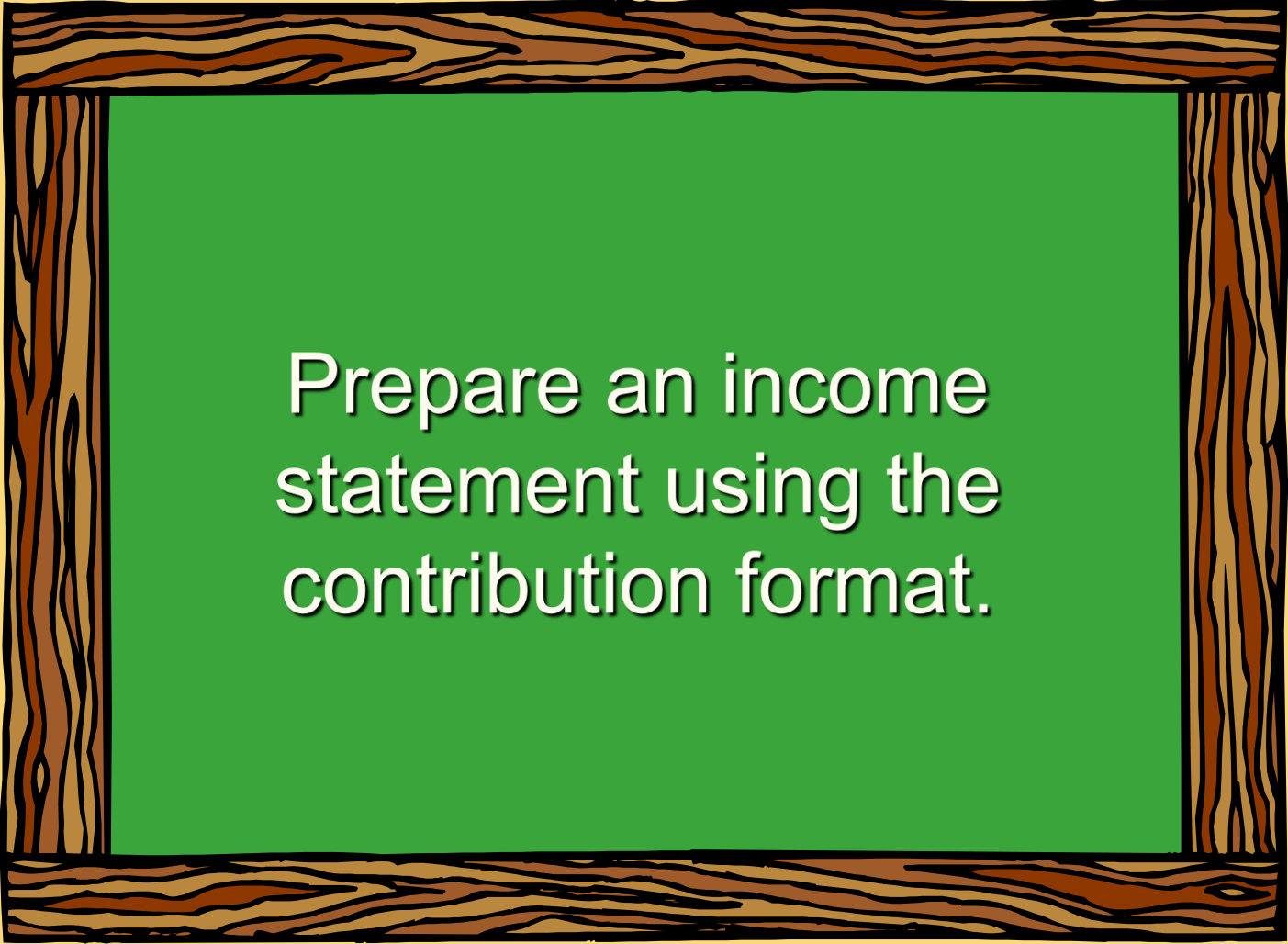
Comparing Results From the Three Methods

The three methods just discussed provide slightly different estimates of the fixed and variable cost components of the mixed cost.

This is to be expected because each method uses differing amounts of the data points to provide estimates.

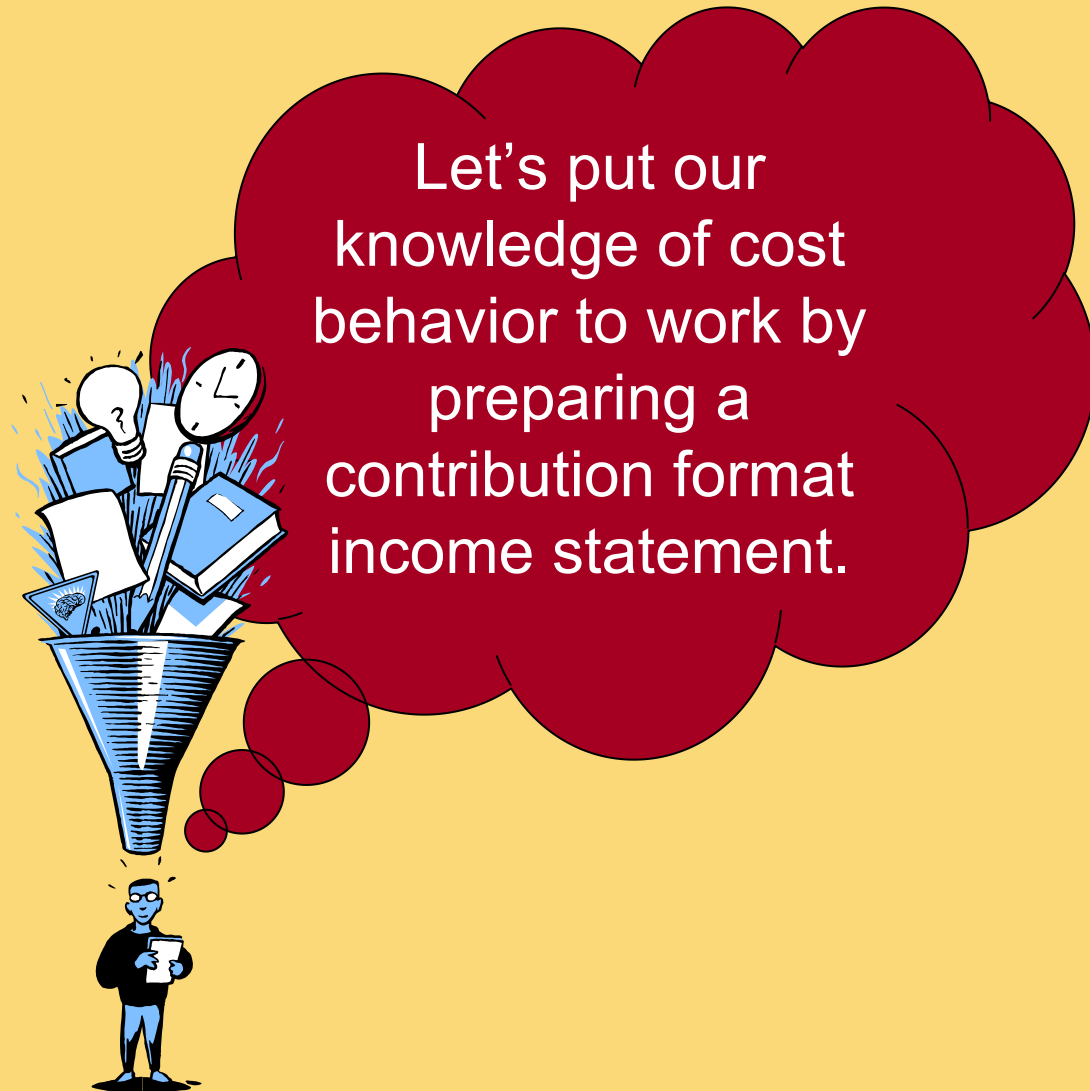
Least-squares regression provides the most accurate estimate because it uses all the data points.

Learning Objective 4



Prepare an income statement using the contribution format.

The Contribution Format



The Contribution Format

	Total	Unit
Sales Revenue	\$ 100,000	\$ 50
Less: Variable costs	60,000	30
Contribution margin	\$ 40,000	\$ 20
Less: Fixed costs	30,000	
Net operating income	<u>\$ 10,000</u>	



The contribution margin format emphasizes cost behavior. Contribution margin covers fixed costs and provides for income.

Uses of the Contribution Format

The contribution income statement format is used as an internal planning and decision making tool.

We will use this approach for:

- 1. Cost-volume-profit analysis (Chapter 6).**
- 2. Budgeting (Chapter 9).**
- 3. Segmented reporting of profit data (Chapter 12).**
- 4. Special decisions such as pricing and make-or-buy analysis (Chapter 13).**

The Contribution Format

Comparison of the Contribution Income Statement with the Traditional Income Statement

Traditional Approach (costs organized by function)		Contribution Approach (costs organized by behavior)	
Sales	\$ 100,000	Sales	\$ 100,000
Less cost of goods sold	<u>70,000</u>	Less variable expenses	<u>60,000</u>
Gross margin	\$ 30,000	Contribution margin	\$ 40,000
Less operating expenses	<u>20,000</u>	Less fixed expenses	<u>30,000</u>
Net operating income	<u><u>\$ 10,000</u></u>	Net operating income	<u><u>\$ 10,000</u></u>

Used primarily for
external reporting.

Used primarily by
management.

A Problem

- Nu Way Inc. manufactures a single product. Selected data from the company's cost records for two recent months are given below:

	Level of Activity	
	July-Low	October-High
Number of units	9000	12000
COGM	\$285000	\$390000
WIP, beginning	\$14000	\$22000
WIP, ending	\$25000	\$15000
DM per unit	\$15	\$15
DL per unit	\$6	\$6
MO total	?	?

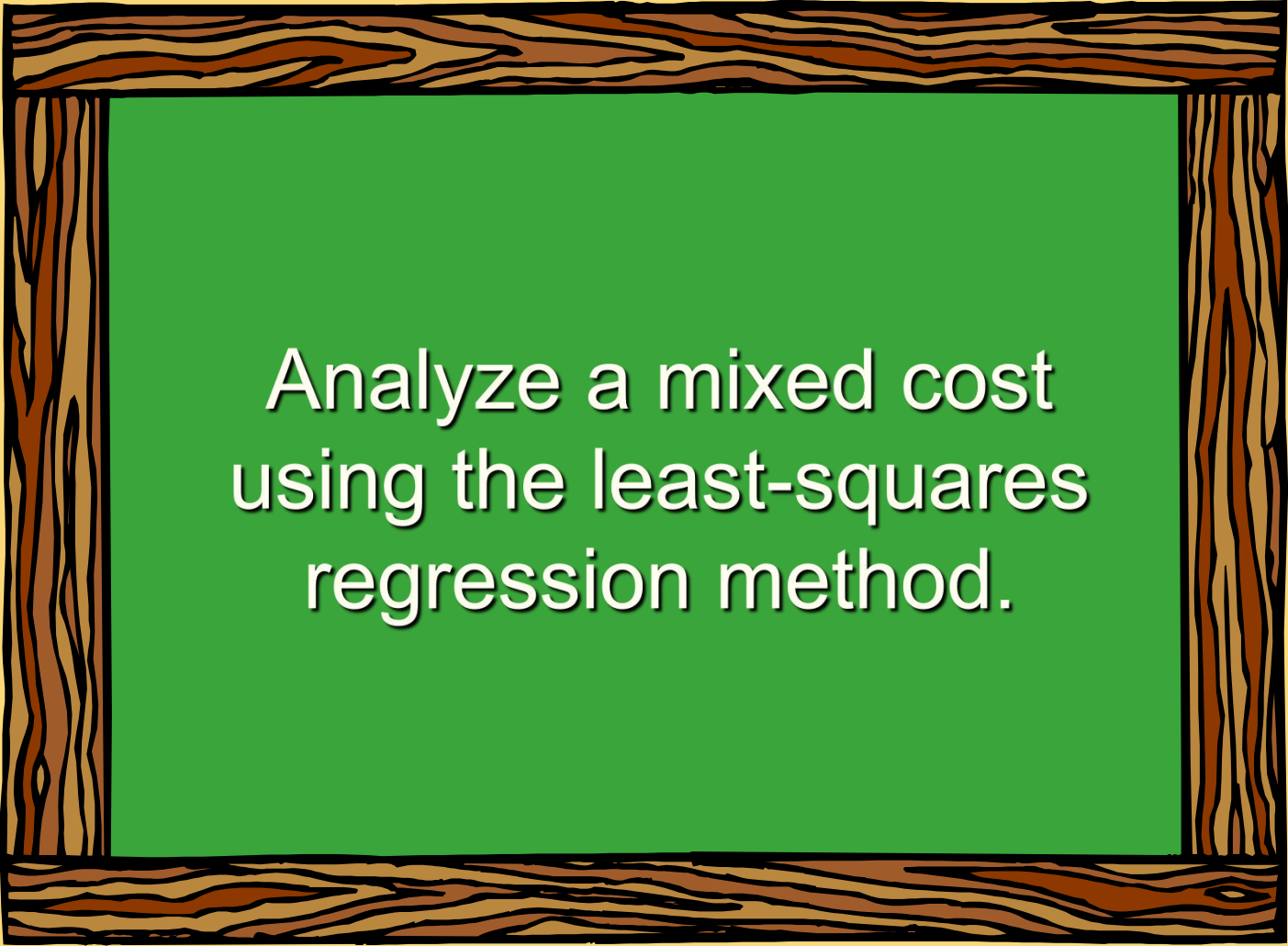
- **The company's MO consists of both variable and fixed cost elements.**
 - ◆ **To estimate the amount of MO added to production. The company had no underapplied or overapplied overhead in either month. (Hint: a schedule of cost of goods manufactured)**
 - ◆ **Use H-L method to estimate the formula for MO.**
 - ◆ **If 9,500 units are produced during a month, what would be the cost of goods manufactured? (Assume that the company's beginning work in process inventory for the month is \$16,000 and that its ending work in process inventory for the month is \$19,000. Also assume that there is no underapplied or overapplied overhead for the month.)**

Least Squares Regression Using Microsoft Excel

Appendix 5A



Learning Objective 5



Analyze a mixed cost
using the least-squares
regression method.

Simple Regression Analysis Example

Matrix, Inc. wants to know its average fixed cost and variable cost per unit.

Using the data to the right, let's see how to do a regression using Microsoft Excel.

	B	C	D
1			
2			
3	Month	Total Cost	Units (Meals)
4	January	\$ 6,720	1,280
5	February	7,260	1,810
6	March	7,270	1,620
7	April	11,060	2,830
8	May	12,580	3,630
9	June	8,660	2,610
10	July	8,580	2,460
11	August	9,550	2,640
12	September	13,050	3,620
13	October	11,060	2,840
14	November	7,320	1,820
15	December	7,370	1,650
16	January	6,790	1,260
17	February	7,480	1,850
18	March	6,990	1,710
19	April	11,400	2,940

Simple Regression Using Excel

You will need three pieces of information from your regression analysis:

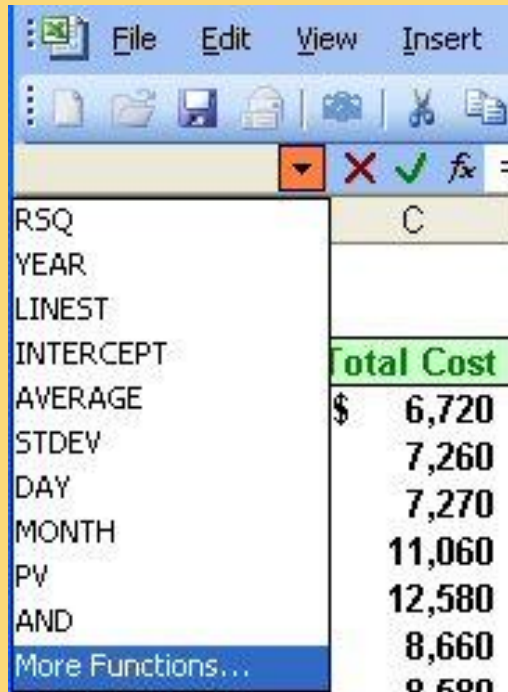
1. Estimated Variable Cost per Unit (line slope)
2. Estimated Fixed Costs (line intercept)
3. Goodness of fit, or R^2

To get these three pieces of information we will need to use *three* different Excel functions.

LINEST, INTERCEPT, & RSQ

	B	C	D
1			
2			
3	Month	Total Cost	Units (Meals)
4	January	\$ 6,720	1,280
5	February	7,260	1,810
6	March	7,270	1,620
7	April	11,060	2,830
8	May	12,580	3,630
9	June	8,660	2,610
10	July	8,580	2,460
11	August	9,550	2,640
12	September	13,050	3,620
13	October	11,060	2,840
14	November	7,320	1,820
15	December	7,370	1,650
16	January	6,790	1,260
17	February	7,480	1,850
18	March	6,990	1,710
19	April	11,400	2,940

Simple Regression Using Excel



Place your cursor in cell F4 and press the = key. Click on the pull down menu and scroll down to “More Functions . . .”

Simple Regression Using Excel

	B	C	D	E	F	G
1						
2						
3	Month	Total Cost	Units (Meals)			
4	January	\$ 6,720	1,280			
5	February	7,260	1,810	=		

The image shows the 'Insert Function' dialog box in Microsoft Excel. The 'Search for a function' section has a text box with the placeholder 'Type a brief description of what you want to do and then click Go' and a 'Go' button. Below this, the 'Or select a category' dropdown is set to 'Statistical'. In the 'Select a function' list, 'SLOPE' is highlighted. Below the list, the function is described as 'SLOPE(known_y's,known_x's)' and 'Returns the slope of the linear regression line through the given data points.' At the bottom, there is a 'Help on this function' link, 'OK' and 'Cancel' buttons.

Search for a function:

Type a brief description of what you want to do and then click Go

Go

Or select a category: Statistical

Select a function:

- QUARTILE
- RANK
- RSQ
- SKEW
- SLOPE**
- SMALL
- STANDARDIZE

SLOPE(known_y's,known_x's)

Returns the slope of the linear regression line through the given data points.

[Help on this function](#)

OK Cancel

Scroll down to the “**Statistical**”, functions. Now scroll down the statistical functions until you highlight “**SLOPE**”

Simple Regression Using Excel

SLOPE =SLOPE(C4:C19,D4:D19)

	B	C	D	E	F	G	H	I
1								
2								
3	Month	Total Cost	Units (Meals)					
4	January	\$ 6,720	1,280					
5	February	7,260	1,810		=SLOPE(C4:C19,D4:D19)			

Function Arguments

SLOPE

Known_y's C4:C19 = {6720;7260;7270;11}

Known_x's D4:D19 = {1280;1810;1620;28}

= 2.768400622

Returns the slope of the linear regression line through the given data points.

1. In the Known_y's box, enter C4:C19 for the range.

2. In the Known_x's box, enter D4:D19 for the range.

Formula result = 2.77

[Help on this function](#)

Simple Regression Using Excel

SLOPE =SLOPE(C4:C19,D4:D19)

	B	C	D	E	F	G	H	I
1								
2								
3	Month	Total Cost	Units (Meals)					
4	January	\$ 6,720	1,280					
5	February	7,260	1,810					

Function Arguments

SLOPE

Known_y's C4:C19 = {6720;7260}

Known_x's D4:D19 = {1280;1810}

= 2.768400625

Returns the slope of the linear regression line through the given data points

Formula result = 2.77

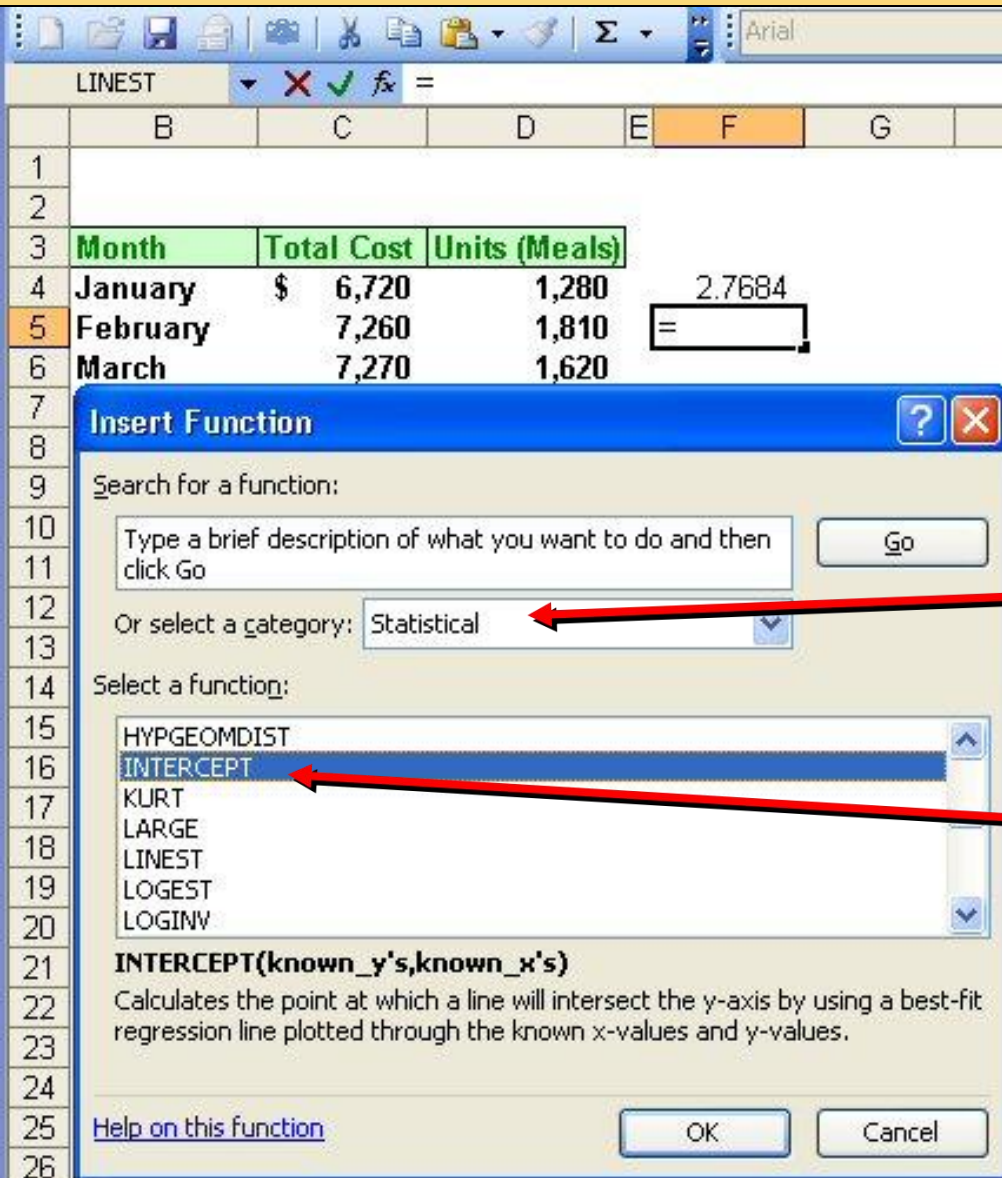
[Help on this function](#)

OK Cancel

Here is the estimate of the slope of the line.

1. In the Known_y's box, enter, C4:C19 for the range.
2. In the Known_x's box, enter, D4:D19 for the range.

Simple Regression Using Excel



Excel spreadsheet showing data for a simple regression analysis. The data table is as follows:

Month	Total Cost	Units (Meals)
January	\$ 6,720	1,280
February	7,260	1,810
March	7,270	1,620

The formula bar shows the LINEST function being entered in cell F5. The 'Insert Function' dialog box is open, showing the 'Statistical' category selected and the 'INTERCEPT' function highlighted in the list. The dialog box also shows the syntax for the INTERCEPT function: **INTERCEPT(known_y's, known_x's)**. Below the syntax, it states: "Calculates the point at which a line will intersect the y-axis by using a best-fit regression line plotted through the known x-values and y-values."

With your cursor in cell F5, press the = key and go to the pull down menu for special functions. Select **Statistical** and scroll down to highlight the **INTERCEPT** function.

Simple Regression Using Excel

INTERCEPT			
=INTERCEPT(C4:C19,D4:D19)			
Month	Total Cost	Units (Meals)	
January	\$ 6,720	1,280	2.7684
February	7,260	1,810	=INTERCEPT(C4:C19,D4:D19)
March	7,270	1,620	
April	11,060	2,830	

Here is the estimate of the fixed costs.

Function Arguments

INTERCEPT

Known_y's C4:C19 = {6720;7260;7270;11}

Known_x's D4:D19 = {1280;1810;1620;2830}

= 2618.724329

Calculates the point at which a line will intersect the y-axis by using a best-fit regression line plotted through the known_y's and known_x's.

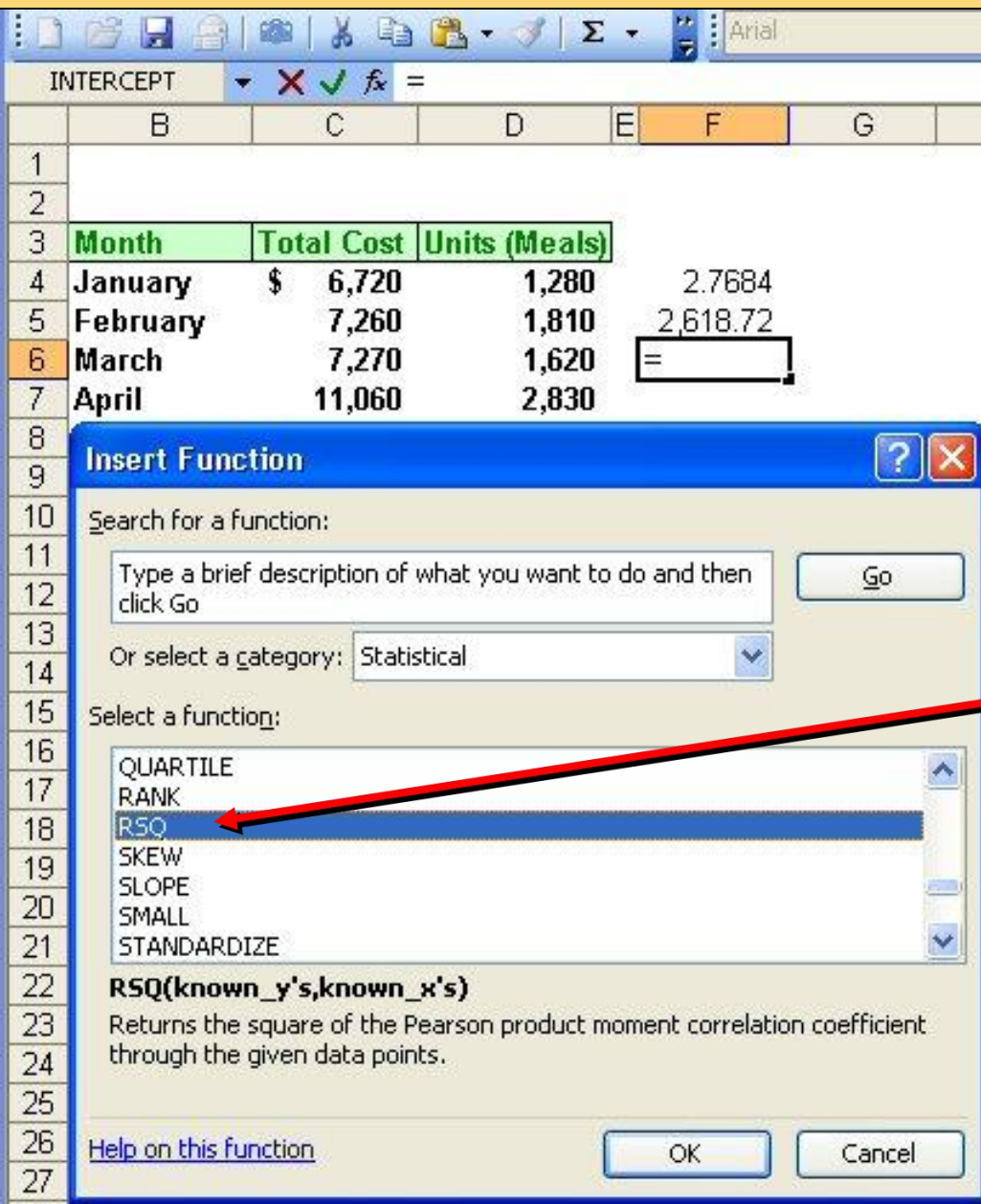
Formulas Result: 2618.724329

[Help on this function](#)

OK Cancel

1. In the Known_y's box, enter C4:C19 for the range.
2. In the Known_x's box, enter D4:D19 for the range.

Simple Regression Using Excel



INTERCEPT

	B	C	D	E	F	G
1						
2						
3	Month	Total Cost	Units (Meals)			
4	January	\$ 6,720	1,280		2.7684	
5	February	7,260	1,810		2,618.72	
6	March	7,270	1,620		=	
7	April	11,060	2,830			

Insert Function

Search for a function:

Type a brief description of what you want to do and then click Go

Or select a category: Statistical

Select a function:

- QUARTILE
- RANK
- RSQ
- SKEW
- SLOPE
- SMALL
- STANDARDIZE

RSQ(known_y's,known_x's)

Returns the square of the Pearson product moment correlation coefficient through the given data points.

[Help on this function](#)

OK Cancel

Finally, we will determine the “*goodness of fit*”, or R^2 , by using the **RSQ** function.

Simple Regression Using Excel

	B	C	D	E	F	G	H
1							
2							
3	Month	Total Cost	Units (Meals)				
4	January	\$ 6,720	1,280		2.7684		
5	February	7,260	1,810		2,618.72		
6	March	7,270	1,620		=RSQ(C4:C19,D4:D19)		
7	April	11,060	2,830				

Here is the estimate of R^2 .

Function Arguments

RSQ

Known_y's: C4:C19 = {6720;7260;7270;11}

Known_x's: D4:D19 = {1280;1810;1620;2830}

= 0.9326524

Returns the square of the Pearson product moment correlation coefficient through the given data points.

Formula result = 0.93

[Help on this function](#)

OK Cancel

1. In the Known_y's box, enter C4:C19 for the range.
2. In the Known_x's box, enter D4:D19 for the range.

End of Chapter 5

